

Yunho Choi

CONTACT INFORMATION	AI/Robotics Researcher Sequor Robotics, Seoul, South Korea	<i>Homepage:</i> ececylh.github.io <i>Mobile:</i> +82-10-2858-4872 <i>Email:</i> ececylh@snu.ac.kr
RESEARCH INTERESTS	Vision-Based Robot Control, Reinforcement Learning, Robotics Foundation Model	
EDUCATION	Ph.D. in Electrical and Computer Engineering • Robot Learning Laboratory, Seoul National University, South Korea • Thesis: Efficient Decision-Making of Embodied Agents Using Local Interest Points • Advisor: Prof. Songhwai Oh	Mar. 2017 - Feb. 2024 GPA: 4.03 / 4.3
	B.S. in Electrical and Computer Engineering • Seoul National University, South Korea	Mar. 2013 - Feb. 2017 GPA: 3.91 / 4.3
PUBLICATIONS	Yunho Choi , Jaeseok Heo, Nuri Kim, and Songhwai Oh, “ KG-Nav: Bridging the Gap between Visual Servoing and Image-Goal Navigation via Keypoint Graph-Based Reinforcement Learning ,” <i>submitted to IEEE Transactions on Robotics (T-RO)</i> . Hwiyeon Yoo , Yunho Choi , Jeongho Park and Songhwai Oh, “ Commonsense-Aware Object Value Graph Navigation for ObjectNav ,” <i>IEEE Robotics and Automation Letters (RA-L)</i> , 2024 and <i>40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)</i> , Sep. 2024. Nuri Kim , Obin Kwon, Hwiyeon Yoo, Yunho Choi , Jeongho Park, and Songhwai Oh, “ Topological Semantic Graph Memory for Image-Goal Navigation ,” in <i>Proc of the Conference on Robot Learning (CoRL)</i> , Dec. 2022. (Oral Presentation, Acceptance Rate: 6.5%) Obin Kwon , Nuri Kim*, Yunho Choi* , Hwiyeon Yoo*, Jeongho Park*, and Songhwai Oh, “ Visual Graph Memory with Unsupervised Representation for Visual Navigation ,” in <i>Proc. of the International Conference on Computer Vision (ICCV)</i> , Oct. 2021. Jaeseok Heo , Yunho Choi , and Songhwai Oh, “Image-Goal Navigation via Metric Mapping and Keypoint based Reinforcement Learning,” in <i>Proc. of the International Conference on Control, Automation and Systems (ICCAS)</i> , Oct. 2021. Yunho Choi and Songhwai Oh, “ Image-Goal Navigation via Keypoint-Based Reinforcement Learning ,” in <i>Proc. of the International Conference on Ubiquitous Robots (UR)</i> , Jul. 2021. Yoonseon Oh , Kyunghoon Cho, Yunho Choi , and Songhwai Oh, “Chance-Constrained Multi-Layered Sampling-Based Path Planning for Temporal Logic-Based Missions,” <i>IEEE Transactions on Automatic Control (TAC)</i> , Dec. 2020. Kyungjae Lee , Jaegu Choy, Yunho Choi , Hogun Kee, and Songhwai Oh, “No-Regret Shannon Entropy Regularized Neural Contextual Bandit Online Learning for Robotic Grasping,” in <i>Proc. of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> , Oct. 2020. Yunho Choi , Nuri Kim, Jeongho Park, Songhwai Oh, “Viewpoint Estimation for Visual Target Navigation by Leveraging Keypoint Detection,” in <i>Proc. of the International Conference on Control, Automation and Systems (ICCAS)</i> , Oct. 2020	

- Nuri Kim, **Yunho Choi**, Minjae Kang, Songhwai Oh, “GOPE: Geometry-Aware Optimal Viewpoint Path Estimation Using a Monocular Camera,” in *Proc. of the International Conference on Control, Automation and Systems (ICCAS)*, Oct. 2020.
- Kyungjae Lee, **Yunho Choi**, and Songhwai Oh, “Inverse Optimal Control from Demonstrations with Mixed Qualities,” in *Proc. of the International Conference on Ubiquitous Robots (UR)*, Jun. 2020.
- Yunho Choi**, Hogun Kee, Kyungjae Lee, Jaegoo Choy, Junhong Min, Sohee Lee, and Songhwai Oh, “**Hierarchical 6-DoF Grasping with Approaching Direction Selection**,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May 2020.
- Kyungjae Lee, **Yunho Choi**, and Songhwai Oh, “Inverse Optimal Control from Demonstrations with Mixed Qualities,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA) Workshop on Learning Legged Locomotion*, May 2019.
- Yunho Choi**, Kyungjae Lee, and Songhwai Oh, “**Distributional Deep Reinforcement Learning with a Mixture of Gaussians**,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May. 2019.
- Hyemin Ahn, Timothy Ha*, **Yunho Choi***, Hwiyeon Yoo*, and Songhwai Oh, “Text2Action: Generative Adversarial Synthesis from Language to Action,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May. 2018.
- Yoonseon Oh, Kyunghoon Cho, **Yunho Choi**, and Songhwai Oh, “Robust Multi-Layered Sampling-Based Path Planning for Temporal Logic-Based Missions,” in *Proc. of the IEEE Conference on Decision and Control (CDC)*, Dec. 2017.
- Yunho Choi**, Inhwon Hwang, and Songhwai Oh, “Wearable Gesture Control of Agile Micro Quadrotors,” in *Proc. of the IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, Nov. 2017.
- Kyunghoon Cho, **Yunho Choi**, and Songhwai Oh, “Reactive Controller Synthesis for UAV Mission Planning,” in *Proc. of the International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, Jun. 2017

HONORS

Awards and Scholarships

- Graduate Scholarship, Korea Foundation for Advanced Studies 2017 - 2024
- Brain Korea 21 Plus Scholarship 2018 - 2022
- Certificate of Excellent Teaching Assistant in Introduction to IoT·AI·Big Data Course, Seoul National University 2018
- Scholarship for Academic Excellence, Seoul National University 2014 -2016
- Summa Cum Laude, Seoul National University 2017
- Samsung Scholarship for Junior Frontier Leader 2012

EXPERIENCE

Research Experience

AI Technology for Guidance of a Mobile Robot to its Goal with Uncertain Maps in Indoor/Outdoor Environments - Ministry of Science and ICT (MSIT)

- Team Lead & Development of a reinforcement learning algorithm for image-goal navigation of a mobile robot Mar. 2019 - 2023

Smart Campus - Samsung Electronics Co., Ltd

- Development of learning-based gesture recognition from sensor data of wearable devices May 2017 - Apr. 2019

Robot Learning from Demonstrations with Mixed Qualities - National Research Foundation (NRF)

- Development of an inverse optimal control algorithm for demonstrations with mixed qualities Mar. 2017 - Feb. 2019

Work Experience

AI Robotics Researcher, Sequor Robotics, Inc.

- Development of robotics foundation models Sep. 2023 - Present

Research Intern at Advanced R&D Group, VD Division, Samsung Electronics Co., Ltd

- Development of a vision-based navigation for an RC car Jul. 2015 - Aug. 2015

Teaching Experience

- Lecturer, Bootcamp for AI Engineers with SOCAR Real-World Data 2021, 2022
- TA, Deep Reinforcement Learning, Seoul National University Spring 2019
- TA, Introduction to Deep Learning, SNU Big Data Academy Fall 2018
- TA, Introduction to IoT·AI·Big Data, Seoul National University Fall 2018
- TA, Introduction to Intelligent Systems, Seoul National University Fall 2017

PROFESSIONAL SERVICES

Reviewer

- IEEE Transactions on Robotics
- IEEE Robotics and Automation Letters
- IEEE International Conference on Robotics and Automation
- IEEE/RSJ International Conference on Intelligent Robots and Systems
- IEEE International Conference on Ubiquitous Robots

PROGRAMMING SKILLS

Language: C++/C, Python, Matlab

Software: ROS, PyTorch, TensorFlow, OpenCV